

Manufacturing & Cost Structure

Overview

Producing a senolytic regenerative therapy (SRT) is a complex biopharmaceutical manufacturing process. It combines small-molecule chemical synthesis (the senolytic compound) with peptide manufacturing (the telomere-stabilizing component) and

aseptic fill-finish (combining, formulating, and packaging the final product).

This document describes the production process, cost structure, and capacity economics that firms face when deciding capex, production volumes, and pricing.

The Manufacturing Process

Step 1: Active Pharmaceutical Ingredient (API) Synthesis

Senolytic compound (small molecule):

- Synthesized via 6–8 step organic chemistry process.
- Key raw materials: proprietary precursor chemicals sourced from 3–4 specialized

suppliers (mostly in Switzerland, Japan, and the US).

- Yield: ~60–70% at commercial scale (improves with process R&D).
- Lead time: 8–12 weeks from raw material to purified API.
- Requires GMP-certified chemical synthesis facility.

Telomere-stabilizing peptide:

- Produced via solid-phase peptide synthesis (SPPS) or recombinant expression in

engineered *E. coli*.

- The peptide is 28 amino acids long, requiring high-purity synthesis.
- Yield: ~40–55% (lower than the small molecule; a key cost driver).
- Lead time: 10–14 weeks.
- Requires specialized peptide manufacturing suite.

Step 2: Formulation

The two APIs are combined with excipients (stabilizers, buffers, cryoprotectants) into the final drug product:

- Gen 1: Lyophilized (freeze-dried) powder in single-use vials, reconstituted at

the clinic with sterile saline before IV infusion.

- Gen 2: Pre-filled syringe with liquid formulation (if subcutaneous delivery

is achieved via R&D).

- Gen 3: Oral tablet with enteric coating and absorption enhancers (requires

significant formulation R&D).

Step 3: Fill-Finish and Packaging

- Aseptic filling in ISO Class 5 cleanroom.
- Each vial/syringe is individually inspected, labeled, serialized.
- Packaged in temperature-controlled shipping containers.
- Shelf life: 24 months at -20°C (Gen 1); future generations target room-temperature stability.

Step 4: Quality Control and Release

- Each batch undergoes 40+ analytical tests (identity, purity, potency, sterility, endotoxin, particle count).
- QC testing takes 4–6 weeks.
- Batch failure rate: ~5–8% for Gen 1 (improves with experience and process R&D).
- Failed batches are destroyed -- total loss of materials and labor.

Cost Structure (Gen 1, per treatment course)

A treatment course = 4 quarterly infusions per year. Costs below are per annual course.

Cost Category	Amount	Notes
Senolytic API	\$3,500	Raw materials + synthesis labor
Peptide API	\$5,200	Most expensive component; low yield
Formulation & fill-finish	\$1,800	Aseptic processing, vials, excipients
Quality control	\$1,200	Testing, batch release, stability
Batch failure allocation	\$900	~7% of batches fail, cost spread
Shipping & cold chain	\$600	-20°C logistics, last-mile to clinic
Regulatory compliance	\$400	Lot tracking, pharmacovigilance
Total COGS per course	\$13,600	At commercial scale (>5,000 courses)

How COGS Varies with Scale

Annual Production (courses)	COGS/course	Notes
500 (minimum viable)	\$22,000	High fixed-cost absorption
2,000	\$17,500	Some scale benefits
5,000	\$13,600	Reference commercial scale
15,000	\$10,800	Bulk purchasing, optimized yields
50,000	\$8,200	World-class efficiency
200,000+ (Gen 3 oral)	\$1,500–3,000	Completely different manufacturing

Key insight: COGS falls significantly with scale, but moving from 5,000 to 50,000 courses/year requires massive capex. Firms must decide how aggressively to build capacity ahead of demand.

COGS by Generation

Generation	COGS/course (at scale)	Key Cost Driver Change
Gen 1	\$13,600	Peptide synthesis yield
Gen 2	\$7,500	Improved peptide yield, simpler formulation
Gen 3	\$2,500	Oral formulation eliminates aseptic processing
Gen 4	\$800	Synthetic biology, commodity-scale

Manufacturing Capacity

What Capacity Means

Capacity is measured in treatment courses per quarter. A firm's capacity determines the maximum number of patients it can supply in any given quarter. Production cannot exceed capacity.

Building Capacity

Facility Type	Capacity (courses/qu)	Build Cost	Build Time	Annual Maintenance
Pilot plant	250	\$25 million	Already exists at st	\$2M/year
Small commercial	1,500	\$120 million	4 quarters	\$8M/year
Standard commercial	5,000	\$350 million	6 quarters	\$20M/year
Large-scale	15,000	\$800 million	8 quarters	\$40M/year
Mega-facility (Gen 3)	50,000	\$500 million	6 quarters	\$25M/year

Notes:

- All firms start with a pilot plant (250 courses/quarter capacity) -- this represents the clinical-trial manufacturing capability converted to commercial use.
 - Capacity expansion is lumpy: you invest the full build cost upfront (as capex), and the capacity comes online after the build time.
 - Capex in the simulation represents quarterly spending on capacity expansion.
- A \$350M facility built over 6 quarters means ~\$58M/quarter in capex if spending evenly. Firms can choose to spend more or less per quarter.
- Capacity depreciates: equipment wears out, requiring reinvestment. Depreciation rate is ~2.5% of gross PPE per quarter (10% annually).
 - Capacity cannot be sold (it is specialized and has no liquid secondary market), but it can be idled (reducing maintenance costs by 50%).

Capacity Utilization and Unit Cost

Running below capacity increases unit cost because fixed costs are spread over fewer units:

Utilization	Effective COGS Multiplier	Reason
>90%	1.0x (optimal)	Full absorption of fixed costs
70–90%	1.1x	Moderate under-absorption
50–70%	1.3x	Significant idle capacity costs
30–50%	1.6x	Severe under-utilization
<30%	2.0x+	Most costs are fixed; very ineffic

Supply Chain Risks

Raw Material Concentration

- The senolytic precursor chemical is sourced from only 3 global suppliers. A supply disruption (factory fire, regulatory action, geopolitical event) could constrain production for 2–4 quarters. Firms can mitigate this by holding safety stock (increases inventory carrying cost).

Cold Chain Fragility

- Gen 1 products must maintain -20°C throughout distribution. A cold-chain break destroys the product. Roughly 2–3% of product is lost in transit (included in COGS above). Gen 3+ products at room temperature eliminate this problem.

Quality Failures

- A serious quality failure (contaminated batch reaching patients) would trigger:
 - Product recall (\$10–50M direct cost depending on scale)
 - FDA clinical hold (1–4 quarters of no sales)
 - Demand shock (patient trust loss)

This is rare (<1% probability per firm per year) but catastrophic.

What This Means for Firm Decisions

- Capex is a long-term bet. Capacity takes 4–8 quarters to build. If you wait for demand to materialize before building, you will miss the window. If you build too early, you burn cash on idle capacity.
- COGS declines with both scale and technology generation. The biggest cost reductions come from advancing to Gen 2 and Gen 3, not from scaling Gen 1.
- Process R&D (distinct from product R&D) reduces COGS. Improving peptide synthesis yields, reducing batch failure rates, and optimizing formulation all lower per-unit cost within a given generation.
- Working capital matters. You must purchase raw materials 8–14 weeks before selling the final product. This means cash is tied up in inventory. A firm growing rapidly needs working capital financing (revolver draws) to bridge the gap.
- Capacity is the hard constraint on revenue. You cannot sell more than you can make. Under-investing in capacity means leaving revenue on the table even if demand is strong.
- The transition from Gen 1 to Gen 3 manufacturing is a paradigm shift. Gen 1 is artisanal biopharmaceutical manufacturing. Gen 3 is closer to mass-market pharmaceutical tablet production. Firms must plan this transition carefully -- Gen 1 facilities cannot be easily converted.